

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

BACKWARD CHAINING

Aim:

To implement the Backward Chaining algorithm in Prolog.

Objective:

To prove a given goal by recursively checking the conditions required to establish it.

Algorithm:

1. **Start.**
2. **Specify the goal.**
3. **Check whether the goal is an existing fact.**
4. **If yes, the goal is proved.**
5. **Otherwise, find a rule whose conclusion matches the goal.**
6. **Treat the rule's conditions as subgoals.**
7. **Recursively prove all subgoals.**
8. **If all subgoals are proved, the goal is proved.**
9. **Otherwise, backtrack and try another rule.**
10. **Stop.**

Flowchart:

START

|

v

Set Goal

|

v

Is Goal a Fact?

/ \

Yes No

| |

Proved Find Rule

| |

| **Rule Found?**

| / \

| **No Yes**

| | |

| **Fail Create Subgoals**

| |

| **Prove Subgoals**

| |

| **All True?**

| / \

	Yes	No
	Proved Backtrack	
\	/	
\	/	
STOP		

Prolog Program:

% Backward Chaining

fact(fever).

fact(cough).

rule(flu, [fever, cough]).

rule(doctor_visit, [flu]).

rule(treatment, [doctor_visit]).

**backward(Goal) :prove(Goal,
[]).**

**prove(Goal, _)
:fact(Goal),
write('Fact found: '),
write(Goal),
nl.**

**prove(Goal, Visited) :-
\+ fact(Goal),
\+ member(Goal, Visited),**

```
rule(Goal, Conditions), write('Trying to prove: '), write(Goal),  
nl,
```

```
prove_all(Conditions, [Goal|Visited]).
```

```
prove_all([], _).
```

```
prove_all([H|T], Visited)  
    :prove(H, Visited),  
    prove_all(T, Visited).
```

Query

```
?- backward(treatment).
```

Output

Trying to prove: treatment

Trying to prove: doctor_visit

Trying to prove: flu

Fact found: fever

Fact found: cough

true.

Another query:

```
?- backward(pneumonia).
```

Output:

```
[1] 99 ?- prove(pneumonia).  
Goal: pneumonia  
Required conditions: [cough,breathlessness]  
Fact found: cough  
Fact found: breathlessness  
[1] 99 ?- prove(pneumonia).
```

Result:

Thus, the Backward Chaining algorithm is successfully implemented to prove a goal by recursively satisfying its required conditions.